

Appendix A: Syntactic and Semantic Case Assignment Diagrams

Appendix A: Syntactic and Semantic Case Assignment Diagrams I

This appendix presents a curated panel of syntactic constituency-style diagrams paired with their categorical pregroup type derivations, covering eight linguistically significant case assignment constructions—from simple intransitive clauses to complex embedded relative clauses with ditransitive verbs. The figure synthesizes the formal correspondences developed throughout the manuscript, making explicit how each surface syntactic pattern maps to a specific morphism composition in the pregroup grammar.

Syntactic Trees and Pregroup Types: Eight Constructions I

The figure below (1) presents eight constructions arrayed in two rows of four panels, each panel containing:

1. **Syntactic tree** (top): a constituency-style diagram with arcs linking argument to predicate, nodes colour-coded by case role following the palette of Linguistic Terms.
2. **Pregroup type formula** (bottom): the formal typing derivation from $??$, showing how Cup contractions collapse all argument wires to sentence type s .

The constructions in 1 span a deliberate difficulty gradient:

Table 1: Appendix A panel index: eight constructions and their categorical complexity indicators.

Panel	Construction	Roles	Type Complexity
1	Intransitive (NOM)	NOM, V	2 boxes, 1 Cup

Syntactic Trees and Pregroup Types: Eight Constructions

II

2	Transitive (NOM+ACC)	NOM, V, ACC	3 boxes, 2 Cups
3	Ditransitive (NOM+DAT+ACC)	NOM, V, DAT, ACC	4 boxes, 3 Cups
4	Passive voice (Patient→NOM)	NOM, V, INS	3 boxes + σ
5	Ergative clause (ERG+ABS)	ERG×2, ABS×2, V	5 boxes, 2 Cups
6	Benefactive (NOM+DAT+ACC+obl+que)	NOM, ACC, DAT	4 boxes, 3 Cups
7	Relative clause (embedded NOM)	NOM×2, V1, V2	6 boxes, 3 Cups

Syntactic Trees and Pregroup Types: Eight Constructions III

8	Causative + Adj + Adv (complex)	NOM, V, ACC, V2, ADV	12 boxes, 5 Cups
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Syntactic Trees and Pregroup Types: Eight Constructions IV

The monotonic increase in Cup count and box count across rows is precisely what ?? captures as the categorical complexity $\kappa(D)$: each additional argument slot requires one additional Cup contraction, and each modifier requires one additional Box–Cup pair.

Ergative Clauses and the Alignment Functor

Panel 5 (Ergative clause) uses the Warlpiri example from ??:
Mariyk-angku (ERG) *yapaku wawirri* (ABS) *parnta-nu* (chased).
The pregroup typing is structurally identical to the transitive (Panel 2), but the morphological realisation is governed by the alignment functor $F_{\text{ERG}} : \mathcal{U} \rightarrow \mathcal{L}_{\text{Warlpiri}}$ from ??, which maps $A \mapsto \text{ERG}$ and $S = P \mapsto \text{ABS}$ rather than $A = S \mapsto \text{NOM}$.

Syntactic Trees and Pregroup Types: Eight Constructions

V

Passivisation as a Swap Morphism

Panel 4 illustrates passivisation: in *Bob is chased by Alice*, the patient Bob is promoted to subject position (NOM) while Alice is demoted to an oblique instrumental (INS). Formally, this is the Swap morphism $\sigma_{A,B} : A \otimes B \rightarrow B \otimes A$ introduced in ???. The pregroup typing includes a σ marker indicating that the type permutation is not a simple contraction sequence but requires an explicit braiding operation.

Syntactic Trees and Pregroup Types: Eight Constructions

VI

Relative Clauses and Wire Threading

Panel 7 (Relative clause) is the most structurally novel: in *The man the dog chased ran*, the head noun *man* is simultaneously the subject of *ran* and the implicit object of *chased* (the gap site). The pregroup type formula $n \cdot n^l \cdot n \cdot (n^r \cdot n \cdot n^l) \cdot (n^r \cdot s) \Rightarrow s$ shows how the relative-clause verb type $n^r \cdot n \cdot n^l$ threads the shared entity wire through two predicate slots — exactly the entity persistence mechanism that DisCoCirc's state wires formalise at discourse level (??).

Syntactic Trees and Pregroup Types: Eight Constructions

VII

Complex Construction and the Complexity Metric

Panel 8 (Causative + Adj + Adv) reaches the highest complexity: 12 boxes and 5 Cup contractions. This corresponds to a categorical complexity value of $\kappa = 12 + 5 = 17$, placing it at the upper end of the single-sentence range plotted in ???. The formula illustrates how clausal complement embedding (the causative taking a VP complement) adds a full additional layer of type nesting beyond even the ditransitive.

Syntactic and Semantic (Categorical) Case Assignment Diagrams
Pregroup Type Derivations across Simple and Complex Constructions

Intransitive (NOM)



$$n \cdot [n' \cdot s] = s$$

Sole argument S assigned NOM;
verb type n' 's contracts with subject n .

Ergative Clause (ERG+ABS)

Transitive (NOM+ACC)



$$n \cdot (n' \cdot s \cdot n') \cdot n = s$$

A=NOM, P=ACC;
verb $n'sn'$ contracts both arguments.

Benefactive (NOM+DAT+ACC+DAT)

Ditransitive (NOM+DAT+ACC)



$$n \cdot (n' \cdot s \cdot n' \cdot n') \cdot n \cdot n = s$$

Three argument slots: subject NOM,
recipient DAT, theme ACC.

Relative Clause (Embedded NOM)

Passive Voice (Patient → NOM)



$$n \cdot (n' \cdot s \cdot n') \cdot n = s$$

Patient promoted to NOM via Swap σ ;
original agent demoted to oblique INS.

Causative + Adj + Adv (Complex)